

Ans to 1

A	B	x	A ⁺	B ⁺	S _A	R _A	S _B	R _B
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	0
0	1	0	1	0	1	0	0	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	0
1	1	0	1	1	X	0	X	0
1	1	1	0	0	0	1	0	1

S R flipflop Excitation Table:

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Kmaps :

(i) for S_A : Bx'

A \ Bx	(00) Bx' Bx'	(01) Bx' Bx	(11) Bx Bx	(10) Bx Bx'
(0)A'				1
(1)A	X	X		X

(ii) for R_A : Bx

A \ Bx	Bx' Bx'	Bx' Bx	Bx Bx	Bx Bx'
A'	X	X	X	
A			1	

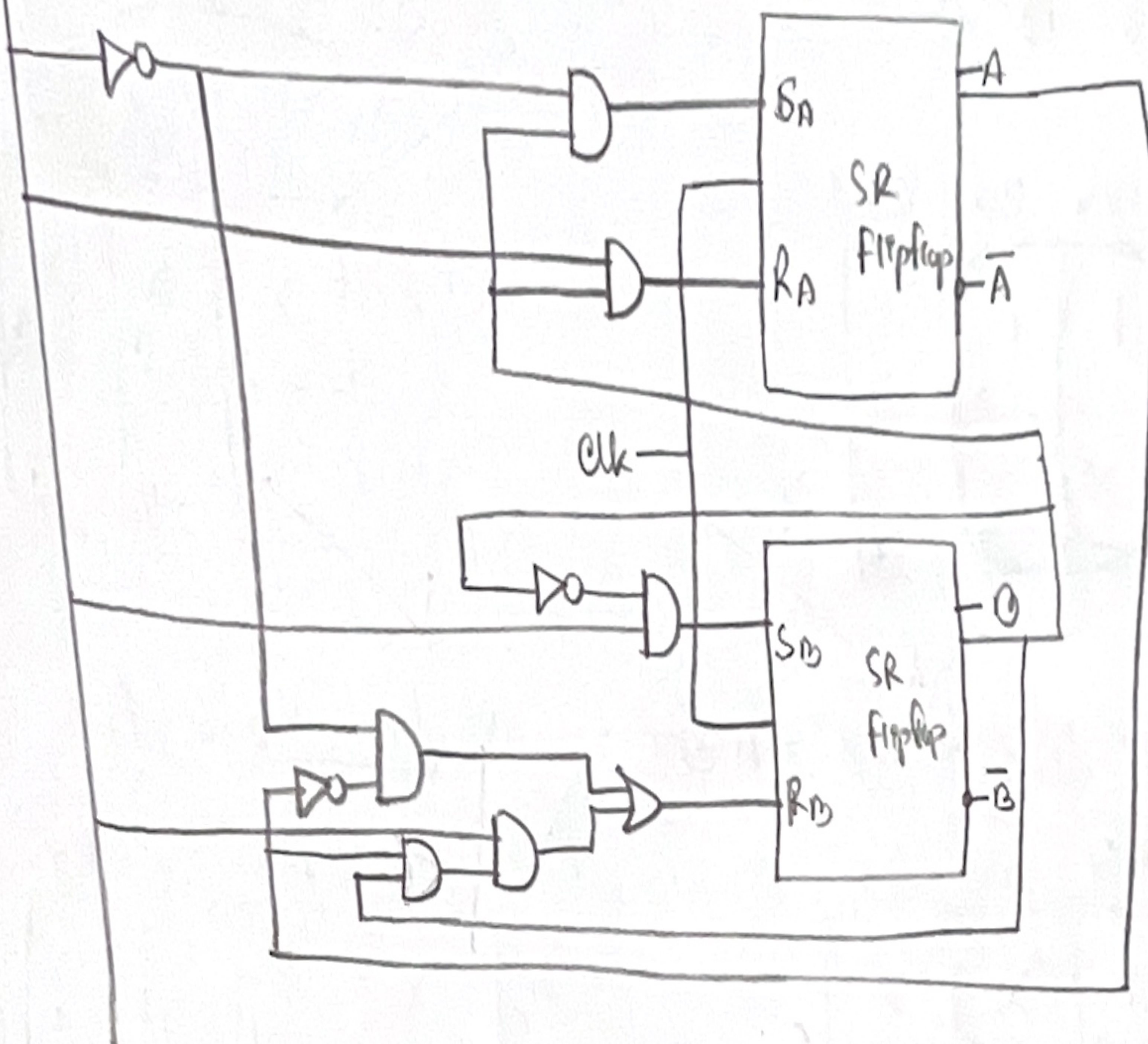
(iii) for S_B : B'x

A \ B'x	B'x' B'x'	B'x' B'x	B'x B'x	B'x B'x'
A'		1	X	
A		1		X

(iv) for R_B : A'x' + ABx

A \ B'x	B'x' B'x'	B'x' B'x	B'x B'x	B'x B'x'
A'	X			1
A	X		1	

x



Ans to 2

State Table :

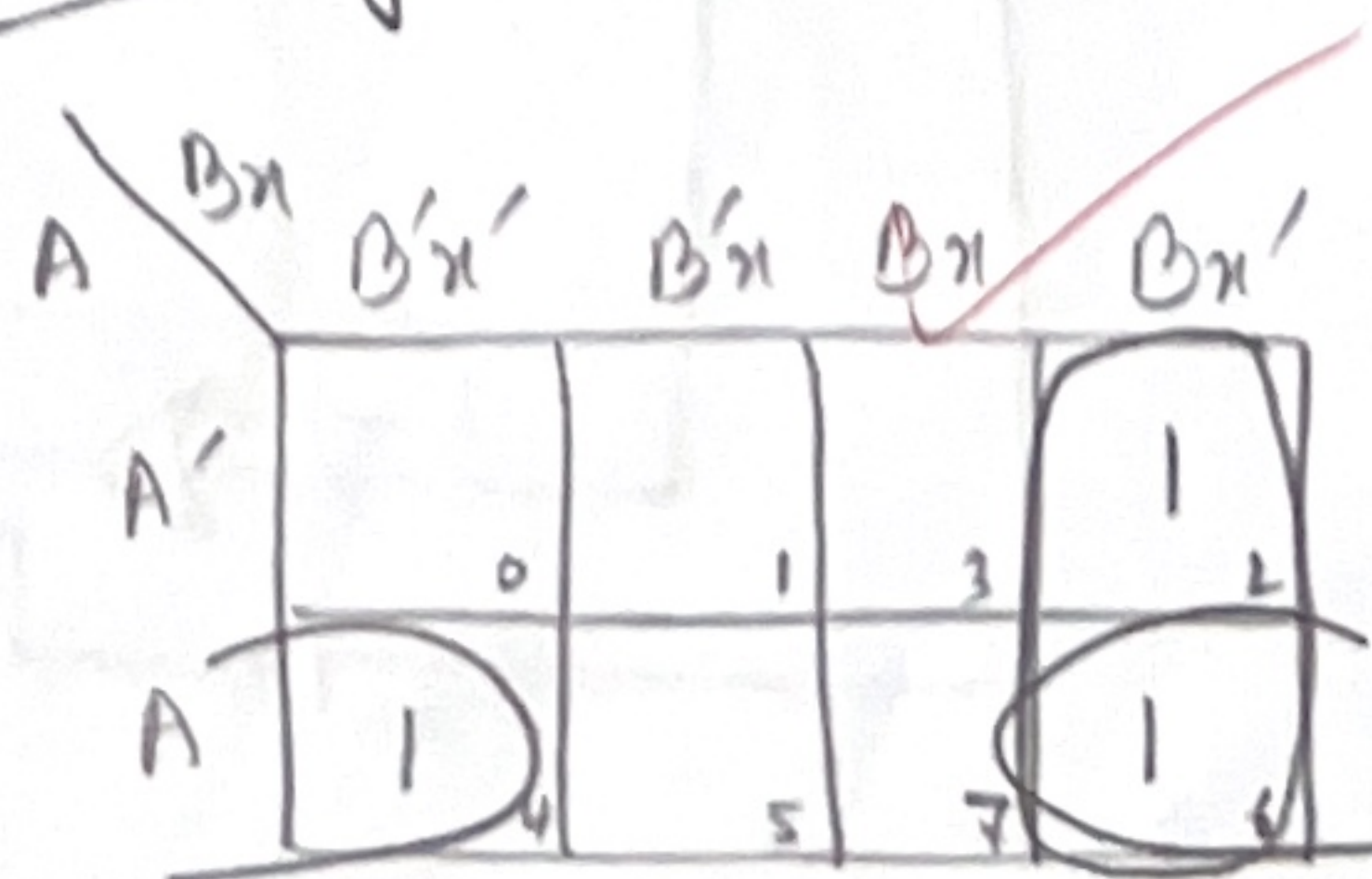
A	B	x	A ⁺	B ⁺	y	J _A	K _A	J _B	K _B
0	0	0	1	0	0	1	X	0	X
0	0	1	1	1	0	1	X	1	X
0	1	0	0	1	1	0	X	X	0
0	1	1	1	1	0	1	X	X	0
1	0	0	1	1	1	X	0	1	X
1	0	1	1	0	0	X	0	0	X
1	1	0	1	1	1	X	0	X	0
1	1	1	1	0	0	X	0	X	1

JK flipflop Excitation table :

Q(d)	Q(d+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Kmaps :

① for y: $Bx' + Ax' = x'(A+B)$



(i) for $J_A \circ B' + x$

A \ Bx	B'x'	B'x	Bx	Bx'
A'	1 0	1 1	1 3	
A	X 4	X 5	X 7	X 6

(ii) for $K_A \circ 0$

A \ Bx	B'x'	B'x	Bx	Bx'
A'	X 0	X 1	X 3	X 2
A				

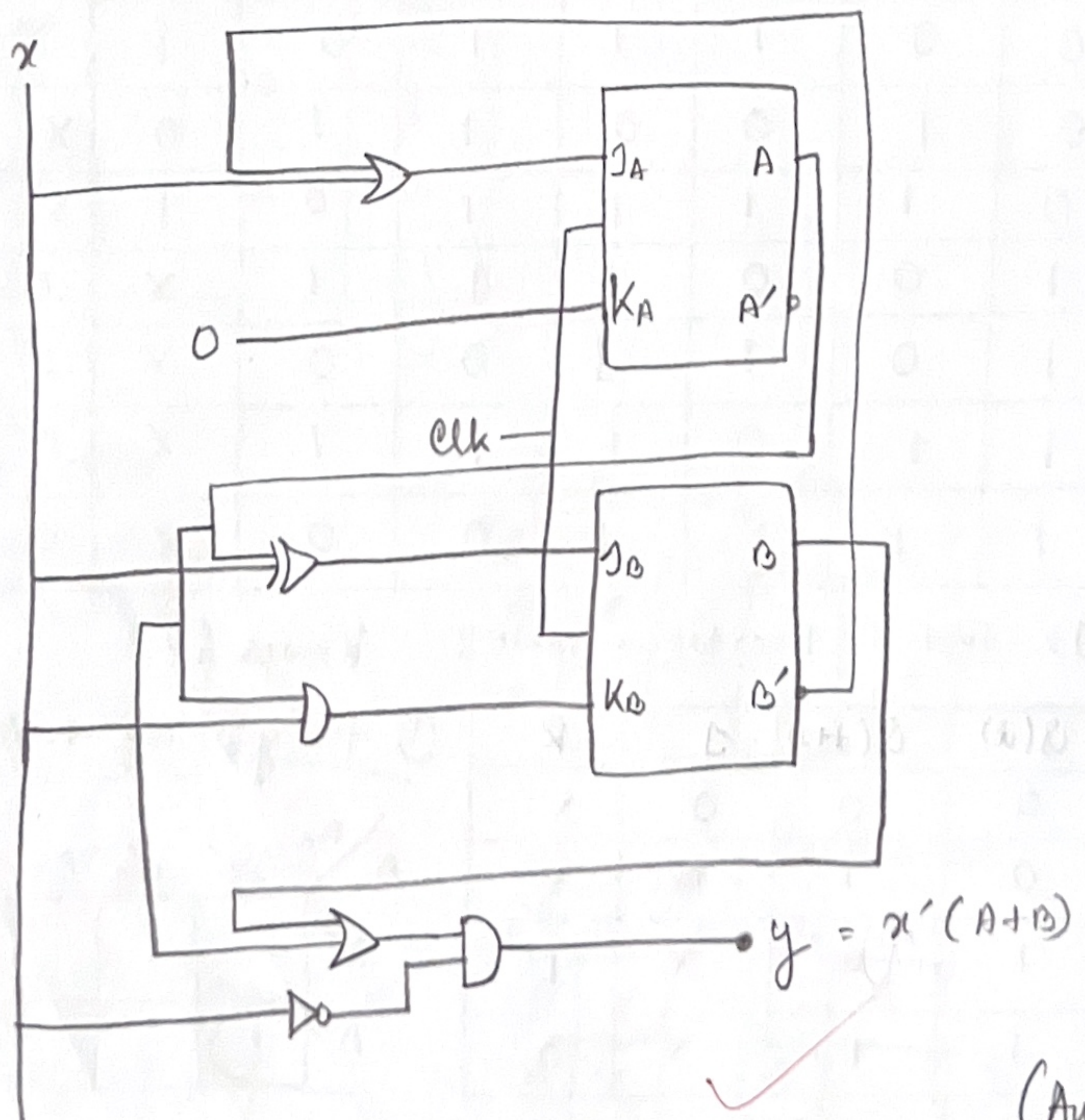
(iv) for $J_B \circ A'x + Ax' = A \oplus x$

A \ Bx	B'x'	B'x	Bx	Bx'
A'		1 1	X 3	X 2
A	1 4		X 7	X 6

(v) for $K_B \circ Ax$

A \ Bx	B'x'	B'x	Bx	Bx'
A'	X 0	X 1		
A	X 4	X 5	1 3	

Circuit Diagram:



(Ans)

Ans to 3

from the given circuit diagram :

Present State: A, B

Next State: A^+, B^+

Input: D_A, D_B, x

Output: y

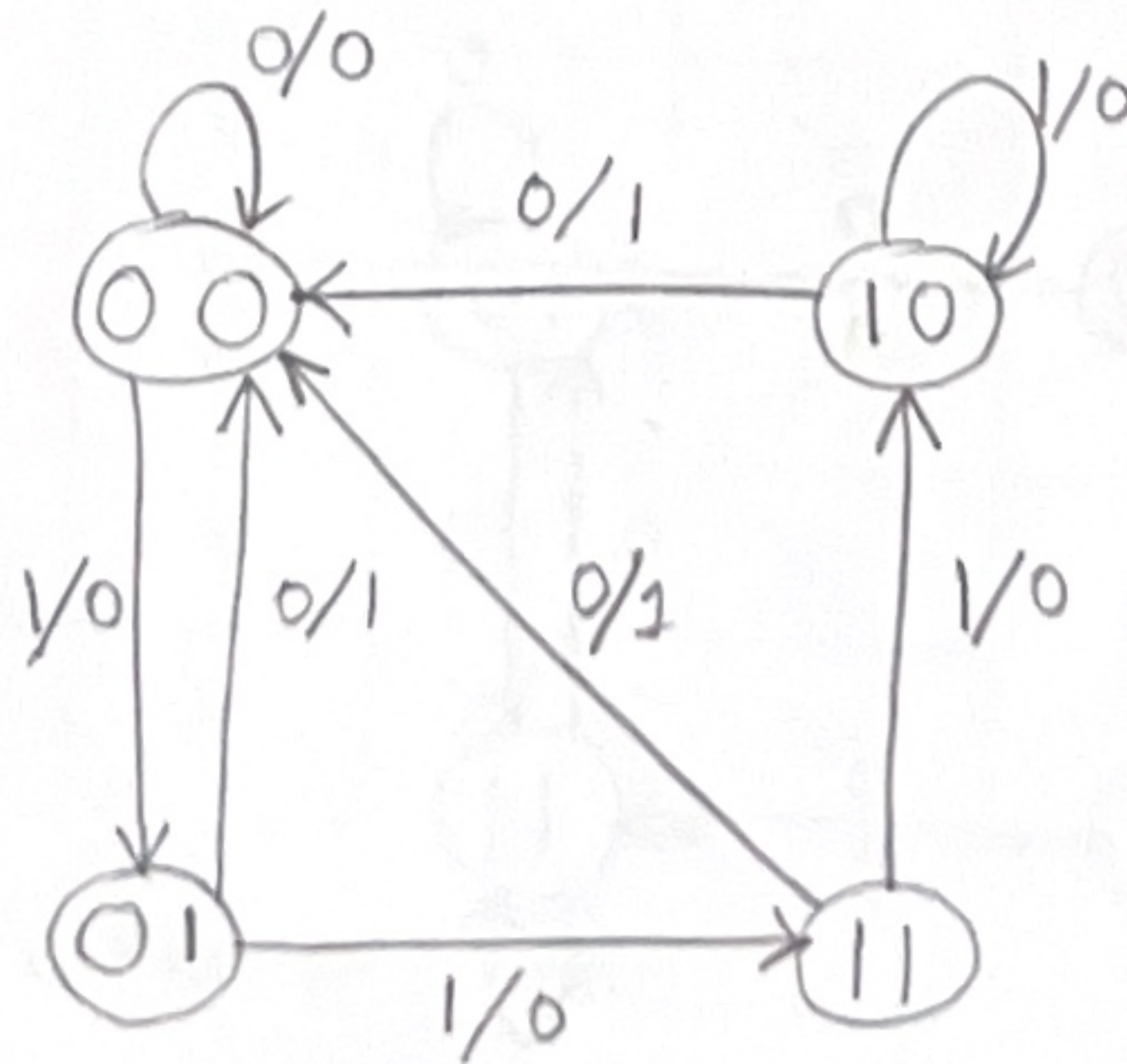
Equation: $D_A = Ax + Bx$
 $D_B = A'x$
 $y = x'(A+B)$

D flip-flop Truth Table:

D	Q	Q'
0	0	1
1	1	0

A	B	x	D_A	D_B	y	A^+	B^+
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	0
1	1	1	1	0	0	1	0

State Diagram:



Ans to 4

From the given circuit diagram:

Present States : A, B

Next States : A^+ , B^+

Input : J_A, K_A, J_B, K_B, x

JK flip-flop Truth Table :

Equation:

$$J_A = A' + B' + x'$$

$$K_A = A'$$

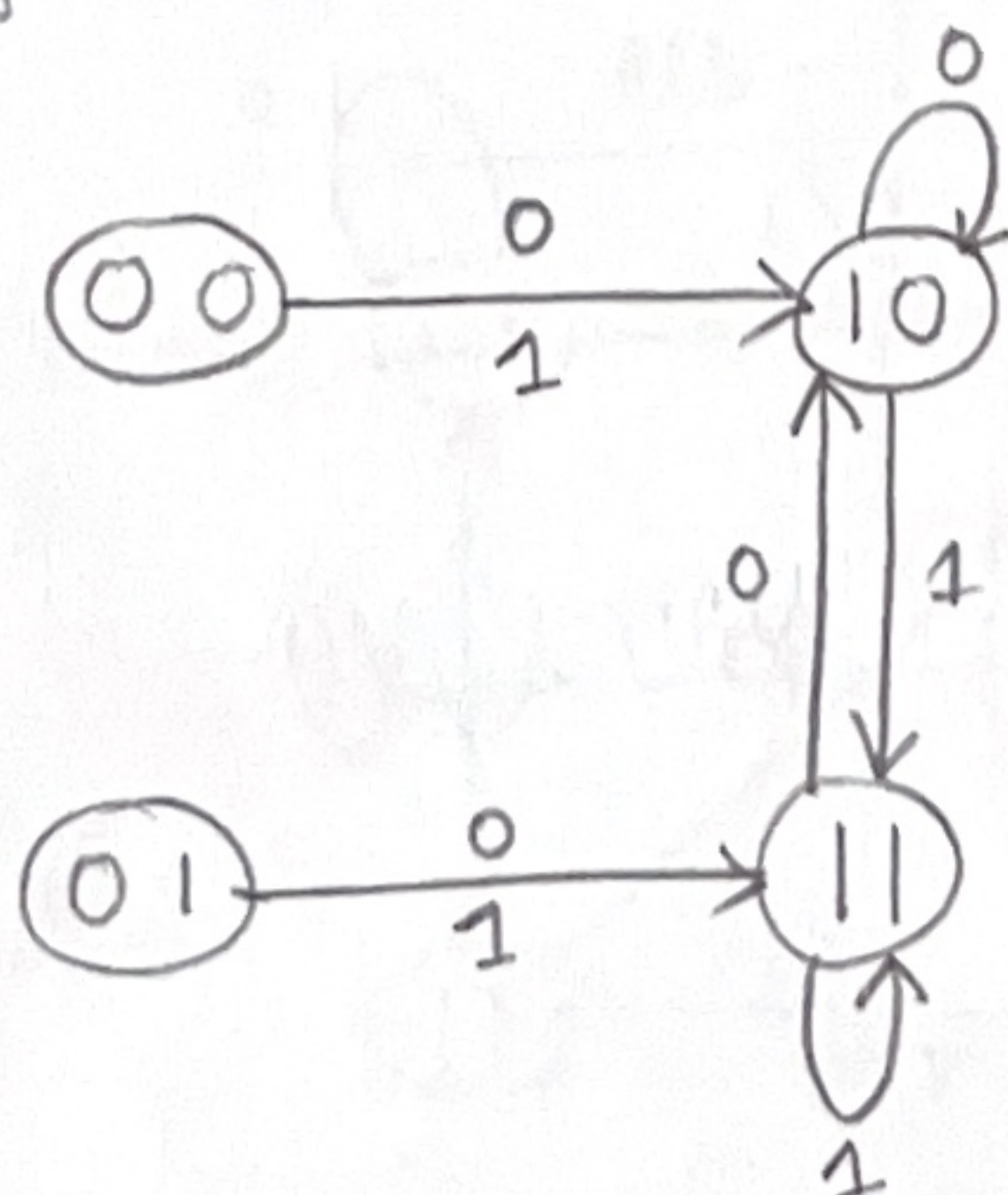
$$J_B = A' \oplus B'$$

$$K_B = Ax'$$

J	K	Q	Q ⁺
0	0	memory	
0	1	0	1
1	0	1	0
1	1	toggle	

A	B	x	J_A	K_A	J_B	K_B	A^+	B^+
0	0	0	1	1	0	0	1	0
0	0	1	1	1	0	0	1	0
0	1	0	1	1	1	0	1	1
0	1	1	1	1	1	0	1	1
1	0	0	1	0	1	1	1	0
1	0	1	1	0	1	0	1	1
1	1	0	1	0	0	1	1	0
1	1	1	0	0	0	0	1	1

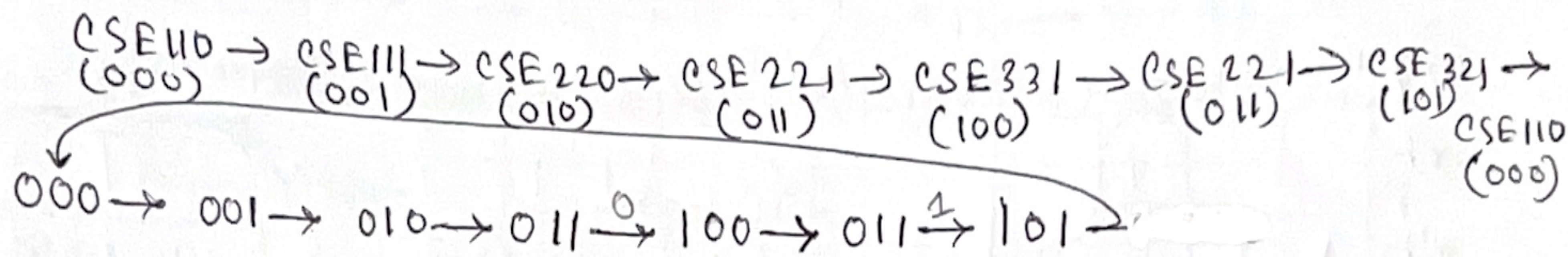
State Diagram:



(Ans)

Ans to 5

det,



x	Present State			Next State			Flipflop inputs		
	A ₂	A ₁	A ₀	A ₂ ⁺	A ₁ ⁺	A ₀ ⁺	T _{A2}	T _{A1}	T _{A0}
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	0	1	1	1	1	1
0	1	0	1	0	0	0	1	0	1
0	1	1	0	x	x	x	x	x	x
0	1	1	1	x	x	x	x	x	x
1	0	0	0	x	x	x	x	x	x
1	0	0	1	x	x	x	x	x	x
1	0	1	0	x	x	x	x	x	x
1	0	1	1	1	0	1	1	1	0
1	1	0	0	x	x	x	x	x	x
1	1	0	1	x	x	x	x	x	x
1	1	1	0	x	x	x	x	x	x
1	1	1	1	x	x	x	x	x	x

T flipflop Excitation Table:

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Kmap :

for T_{A2} :

$\backslash A_1 A_0$	$A_1 A_0$	$A_1' A_0'$	$A_1' A_0$	$A_1 A_0'$
$\backslash A_2$				
$\backslash A_2'$			1	
$\backslash A_2$	1	1	X	X
$\backslash A_2'$	X	X	X	X
$\backslash A_2$	X	X	1	X

for T_{A1} :

$\backslash A_1 A_0$	$A_1 A_0$	$A_1' A_0'$	$A_1' A_0$	$A_1 A_0'$
$\backslash A_2$				
$\backslash A_2'$		1	1	
$\backslash A_2$	1		X	X
$\backslash A_2'$	X	X	X	X
$\backslash A_2$	X	X	1	X

for T_{A0} :

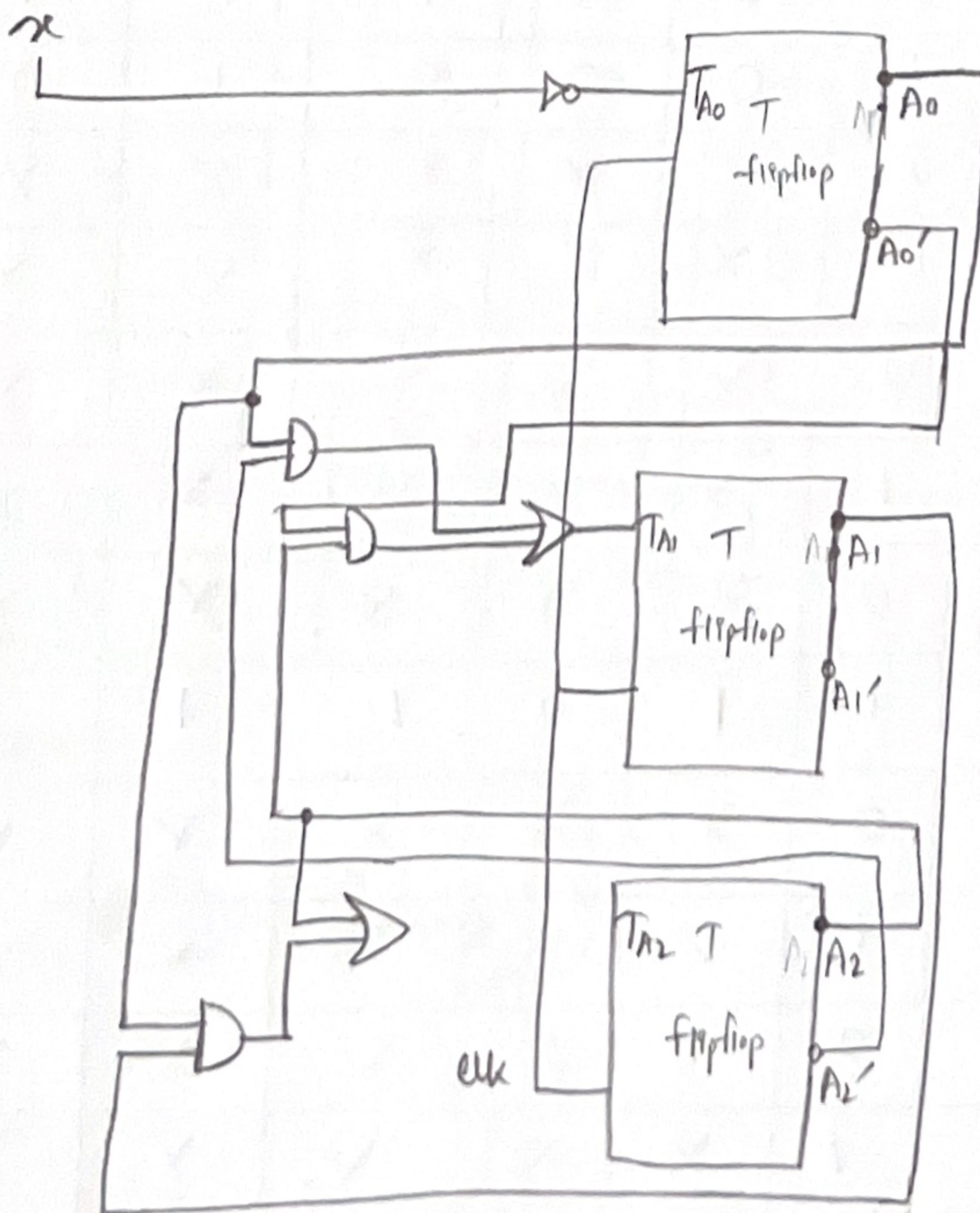
$\backslash A_1 A_0$	$A_1 A_0$	$A_1' A_0'$	$A_1' A_0$	$A_1 A_0'$
$\backslash A_2$				
$\backslash A_2'$	1	1	1	
$\backslash A_2$	1	1	X	
$\backslash A_2'$	X	X	X	
$\backslash A_2$	X	X		

$$T_{A2} = A_2 + A_1 A_0$$

$$T_{A1} = A_2' A_0 + A_2 A_0'$$

$$T_{A0} = X'$$

Circuit Diagram :



Ans to 6

given, $3 \rightarrow 4 \rightarrow 6 \rightarrow 10 \rightarrow 12 \rightarrow 13 \rightarrow 15 \rightarrow 3$
 $(0011) \rightarrow (0100) \rightarrow (0110) \rightarrow (1010) \rightarrow (1100) \rightarrow (1101) \rightarrow (1111) \rightarrow (0011)$

Present State				Next State				FlipFlop Inputs			
A_3	A_2	A_1	A_0	A_3^+	A_2^+	A_1^+	A_0^+	D_{A3}	D_{A2}	D_{A1}	D_{A0}
0	0	0	0	0	0	1	1	0	0	1	1
0	0	0	1	0	0	1	1	0	0	1	1
0	0	1	0	0	0	1	1	0	0	1	1
0	0	1	1	0	1	0	0	0	1	0	0
0	1	0	0	0	1	1	0	0	1	1	0
0	1	0	1	0	0	1	1	0	0	1	1
0	1	1	0	1	0	1	0	1	0	1	0
0	1	1	1	0	0	1	1	0	0	1	1
1	0	0	0	0	0	1	1	0	0	1	1
1	0	0	1	0	0	1	1	0	0	1	1
1	0	1	0	1	1	0	0	1	1	0	0
1	0	1	1	0	0	1	1	0	0	1	1
1	1	0	0	1	1	0	1	1	1	0	1
1	1	0	1	1	1	1	1	1	1	1	1
1	1	1	0	0	0	1	1	0	0	1	1
1	1	1	1	0	0	1	1	0	0	1	1

D flipflop Excitation Table:

K map:

$Q(t)$	$Q(t+1)$	D
0	0	0
0	1	1
1	0	0
1	1	1

① - for D_{A3} :

$A_3 A_2$ \ $A_1 A_0$	$A_1 A_0$	$A_1 A_0$	$A_1 A_0$	$A_1 A_0$
$A_3 A_2$	$A_1 A_0$	$A_1 A_0$	$A_1 A_0$	$A_1 A_0$
$A_3 A_2$				
$A_3 A_2$				1
$A_3 A_2$	1	1		
$A_3 A_2$				1

$$D_{A3} = A_3 A_2 A_1' + A_3' A_2 A_1 A_0' + A_3 A_2' A_1 A_0'$$

(ii) For DA_2 :

A_3A_2	A_1A_0	A_1A_0	A_1A_0	A_1A_0
A_3A_2				
A_3A_2				
A_3A_2				
A_3A_2				

$$DA_2 = A_2 A_1 A_0' + A_3 A_2 A_1' + A_3 A_2' A_1 A_0 + A_3 A_2' A_1 A_0'$$

(iii) For DA_1 :

A_3A_2	A_1A_0	A_1A_0	A_1A_0	A_1A_0
A_3A_2				
A_3A_2				
A_3A_2				
A_3A_2				

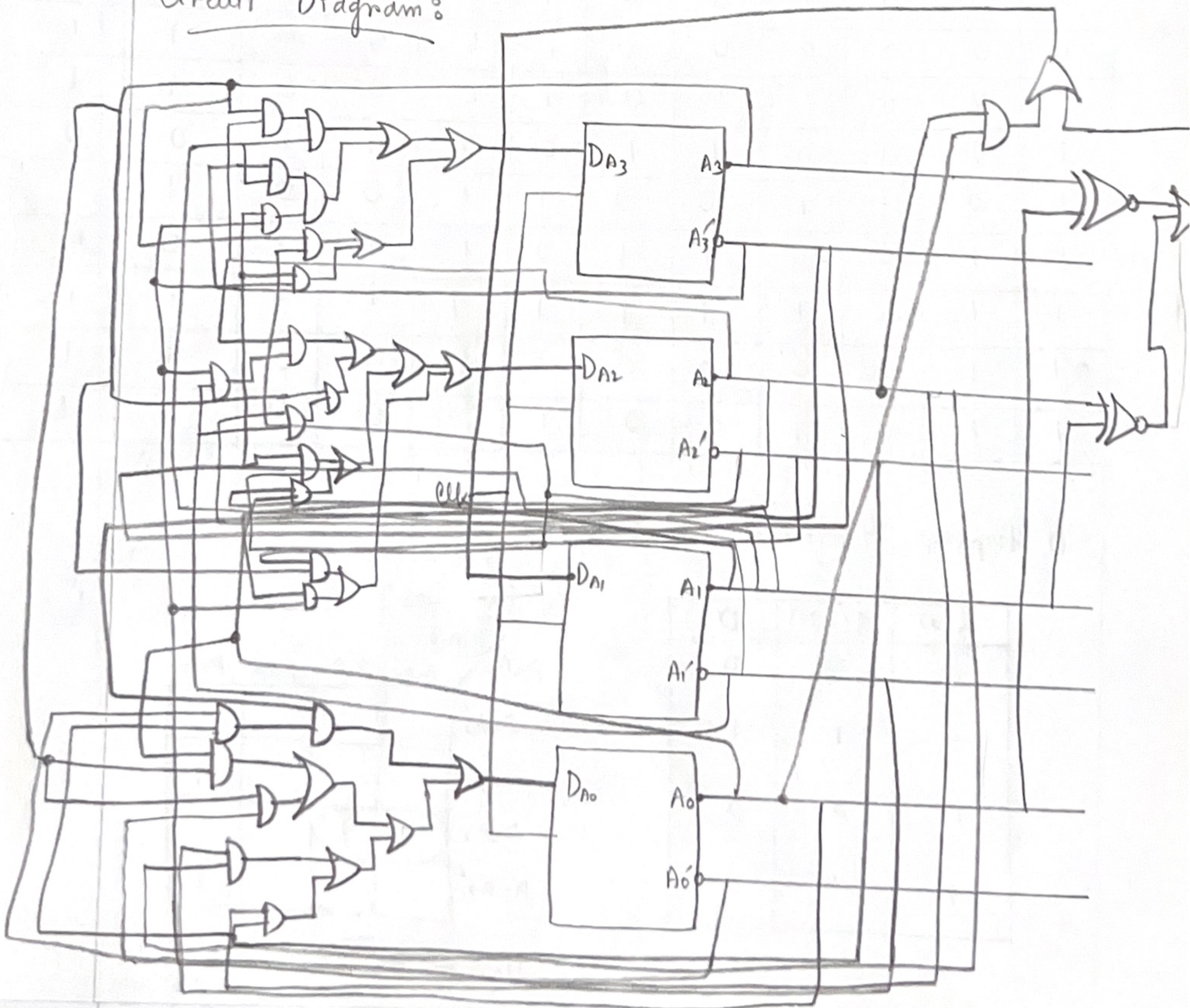
$$DA_1 = A_2 A_1' + A_3 A_0' + A_2 A_0 + A_2 A_1 + A_3 A_0 = (A_2 \odot A_1) + (A_3 \odot A_0) + A_2 A_0$$

(iv) For DA_0 :

A_3A_2	A_1A_0	A_1A_0	A_1A_0	A_1A_0
A_3A_2				
A_3A_2				
A_3A_2				
A_3A_2				

$$DA_0 = A_2' A_1' + A_2 A_3 A_0 + A_3 A_2' A_0'$$

Circuit Diagram:

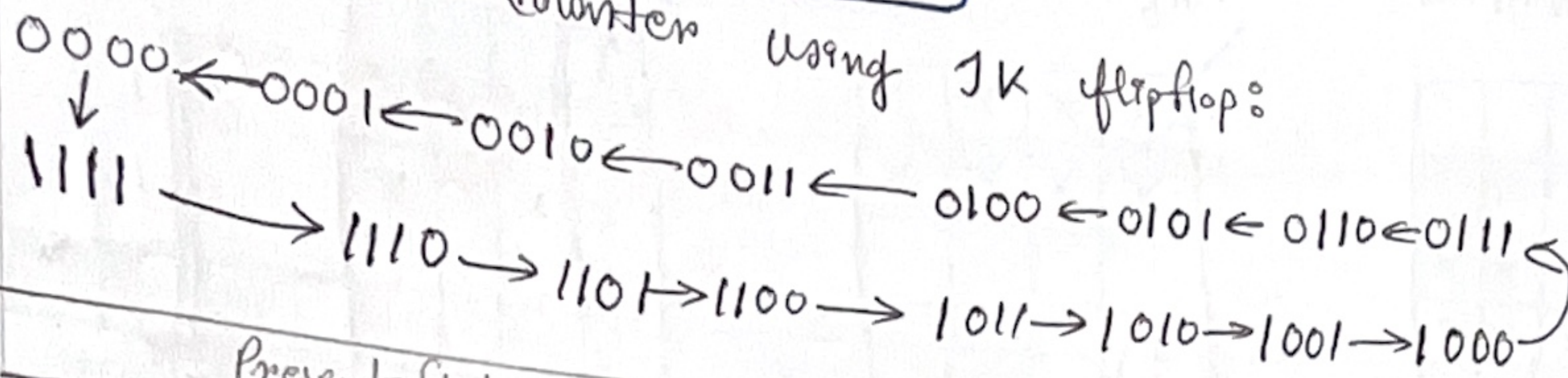


4 bit down

counter

Ans to 7

using JK flipflop:



Present State				Next State				Flip-flop input							
A ₃	A ₂	A ₁	A ₀	A ₃	A ₂	A ₁	A ₀	J _{A₃}	K _{A₃}	J _{A₂}	K _{A₂}	J _{A₁}	K _{A₁}	J _{A₀}	K _{A₀}
0	0	0	0	1	1	1	1	1	X	1	X	1	X	1	X
0	0	0	1	0	0	0	0	0	X	0	X	0	X	X	1
0	0	1	0	0	0	0	1	0	X	0	X	X	1	1	X
0	0	1	1	0	0	1	0	0	X	0	X	X	0	X	1
0	1	0	0	0	0	1	1	0	X	X	1	1	X	1	X
0	1	0	1	0	1	0	0	0	X	X	0	0	X	X	1
0	1	1	0	0	1	0	1	0	X	X	0	X	1	1	X
0	1	1	1	0	1	1	0	0	X	X	0	X	0	X	1
1	0	0	0	0	1	1	1	X	1	1	X	1	X	1	X
1	0	0	1	1	0	0	0	X	0	0	X	0	X	X	1
1	0	1	0	1	0	0	1	X	0	0	X	X	1	1	X
1	0	1	1	1	0	1	0	X	0	0	X	X	0	X	1
1	1	0	0	1	0	1	1	X	0	X	1	1	X	1	X
1	1	0	1	1	1	0	0	X	0	X	0	0	X	X	1
1	1	1	0	1	1	0	1	X	0	X	0	X	1	1	X
1	1	1	1	1	1	1	0	X	0	X	0	X	0	X	1

JK flipflop Excitation Table :

Q(d)	Q(d+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$J_{A_3}^0 \ A_2' A_1' A_0'$

1			
X	X	X	X
X	X	X	X

$K_{A_3}^0 \ A_2' A_1' A_0'$

X	X	X	X
X	X	X	X
1			

$J_{A_2}^0 \ A_2' A_0'$

1			
X	X	X	X
X	X	X	X

$K_{A_2}^0$

X	X	X	
1			
1			
X	X	X	

$J_{A_1}^0 \ A_0'$

1		X	X
1		X	X
1		X	X
1		X	X

$K_{A_1}^0 \ A_0'$

X	X		1
X	X		1
X	X		1
X	X		1

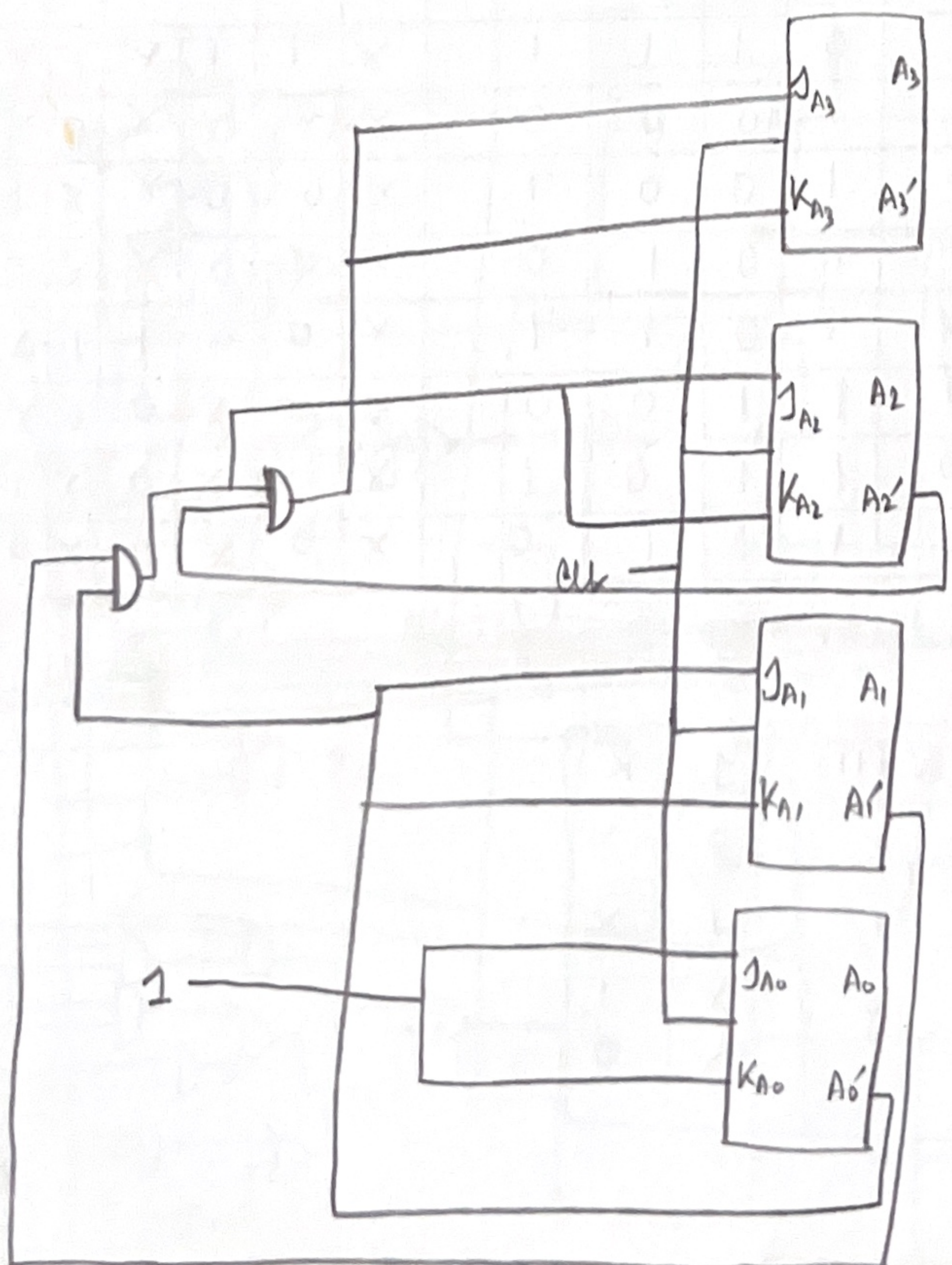
$J_{A_0}^0 \ 1$

1	X	X	1
1	X	X	1
1	X	X	1
1	X	X	1

$K_{A_0}^0 \ 1$

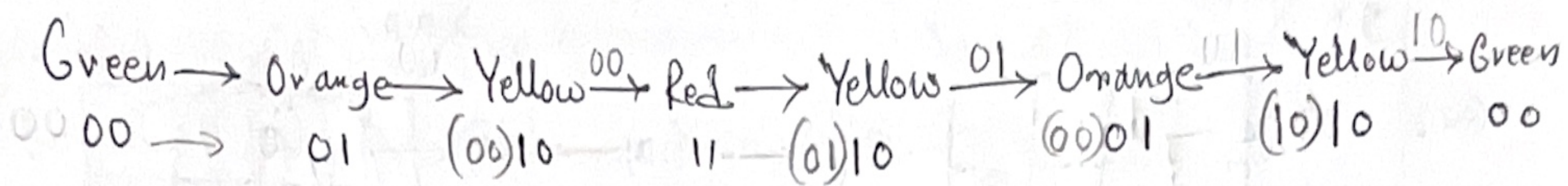
X	1	1	
X	1	1	
X	1	1	
X	1	1	

Circuit Diagram:



Ans to 8

given:

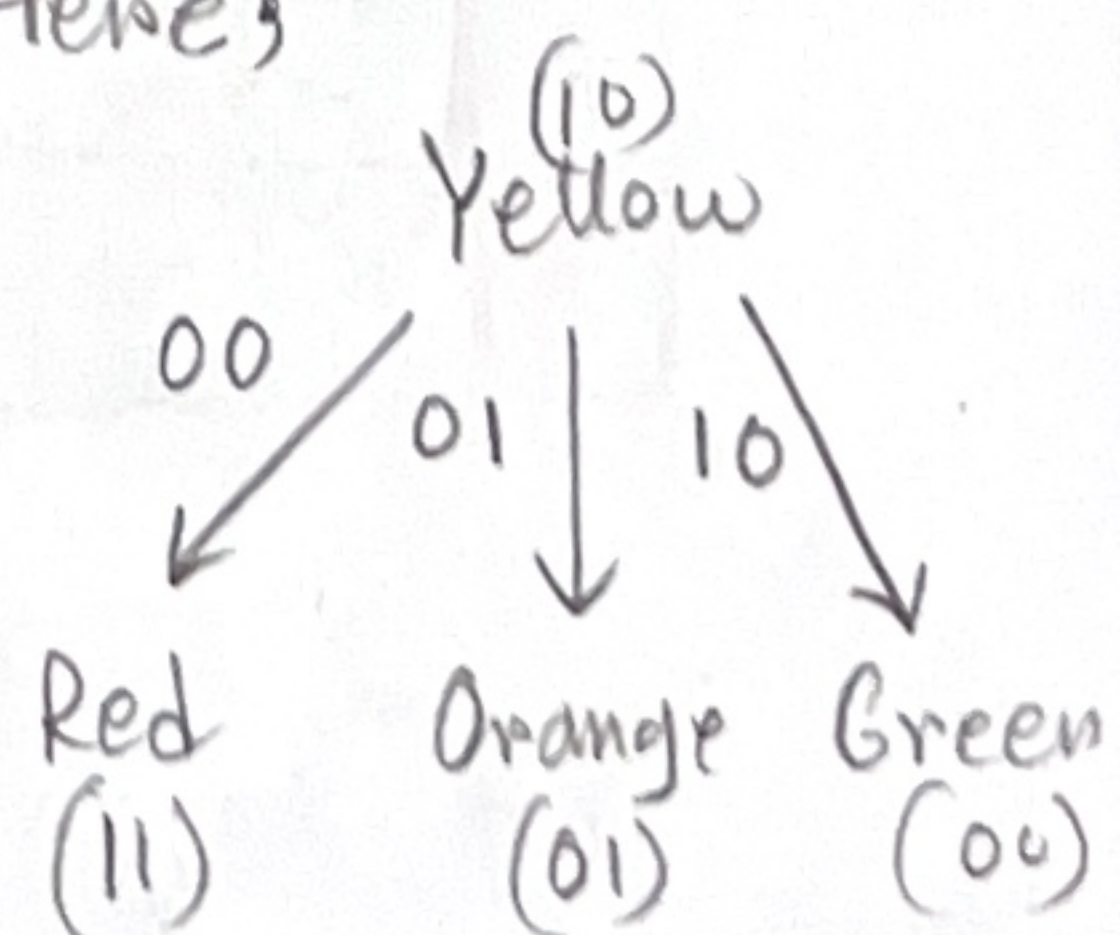


External Input		Present State		Next State				Flipflop Input	
y	x	A ₁	A ₀			A ₁ ⁺	A ₀ ⁺	T _{A₁}	T _{A₀}
0	0	0	0			0	1	0	1
0	0	0	1			1	0	1	1
0	0	1	0			1	1	0	1
0	0	1	1			1	0	0	1
0	1	0	0			0	1	0	1
0	1	0	1			1	0	1	1
0	1	1	0			0	1	1	1
0	1	1	1			1	0	0	1
1	0	0	0			0	1	0	1
1	0	0	1			1	0	1	1
1	0	1	0			0	0	1	0
1	0	1	1			1	0	0	1
1	1	0	0			0	1	0	1
1	1	0	1			1	0	1	1
1	1	1	0			X	X	X	X
1	1	1	1			1	0	0	1

T Flipflop Excitation Table:

Q(n)	Q(n+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Hence,



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for T_{A0} :

for T_{A1} :

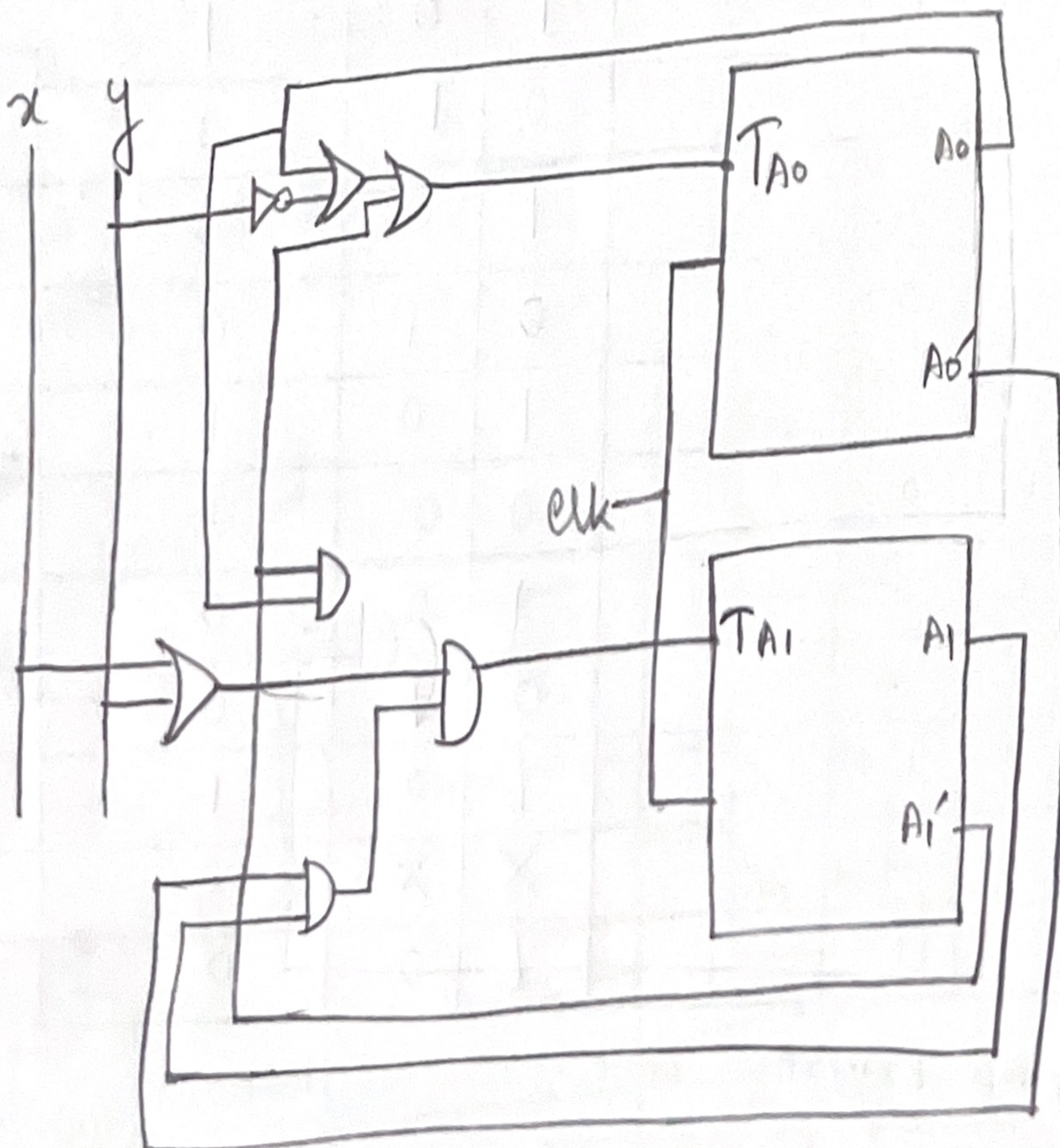
$y \ x$	$A_1 A_0$ $A_1' A_0'$	$A_1' A_0$	$A_1 A_0'$
$y' x'$	1		
$y' x$	1		1
$y x$	1		X
$y x'$	1		1

$y \ x$	$A_1 A_0$ $A_1' A_0'$	$A_1' A_0$	$A_1 A_0'$
$y' x'$	1	1	1
$y' x$	1	1	1
$y x$	1	1	X
$y x'$	1	1	

$$T_{A1} = A_1 A_0 + A_1 A_0' (x + y)$$

$$T_{A0} = y' + A_1' + A_0$$

Circuit Diagram :



Ans to 9

given,

Green $\xrightarrow{0}$ Yellow $\xrightarrow{1}$ Red $\xrightarrow{0}$ Yellow $\xrightarrow{1}$ Green
 00 (0)01 10 (1)01 00

x	Present State			Next State		Flip-flop Input			
	A ₁	A ₀		A ₁ ⁺	A ₀ ⁺	J _{A₁}	K _{A₁}	J _{A₀}	K _{A₀}
0	0	0		0	1	0	X	1	X
0	0	1		1	0	1	X	X	1
0	1	0		0	1	X	1	1	X
0	1	1		X	X	X	X	X	X
1	0	0		0	1	0	X	1	X
1	0	1		0	0	0	X	X	1
1	1	0		0	1	X	1	1	X
1	1	1		X	X	X	X	X	X

JK FlipFlop Excitation Table :

Q(d)	Q(d+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

For J_{A₁} : $x'A_0$

x	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x'	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x	1	X	X	X
x'			X	X

For J_{A₀} : 1

x	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x'	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x	1	X	X	1
x'	1	X	X	1

For K_{A₁} : 1

x	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x'	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x	X	X	X	1
x'	X	X	X	1

For K_{A₀} : 1

x	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x'	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀	A ₁ A ₀
x	X	1	X	X
x'	X	1	X	X

Circuit Diagram :

